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Digital image correlation based characterization of rubber material at large shear deformations in an extended temperature range

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Abstract

This contribution deals with a new shear device, which is applied for rubber measurements in simple shear in an extended temperature range. As a special feature for the loading system, no bonding technique, or vulcanization is used, but a form fit connection via pins. The measured stress–shear value curves for different temperature levels are evaluated with a standard method from GOM ARAMIS. In addition, a new approximation based evaluation method is introduced and applied for shear value determination including smoothing of the raw digital image correlation data with consideration of characteristic noise properties. This enables the analysis of noise induced error influences. It was proved by the approximation based method, that the standard ARAMIS evaluation provides shear value results, which are suitable for performed characterization tests. As a consequence, the nearly homogeneous measurement data can be used for parameter fitting of constitutive rubber models.

KEYWORDS

characterization, digital image correlation, EPDM rubber, rubber phenomenology, shear device, shear measurements, smoothing

1 | INTRODUCTION

Simple shear of rubber is an important deformation state for many technical applications, for example, chassis bushings and dampers (cf. [13,21]). The basic problem here is to apply the load uniformly in order to achieve a nearly homogeneous deformation state (cf. [26]). Note that an inhomogeneous deformation state can lead to errors in parameter identification (cf. [2]). In addition, simple shear in technical applications is often overlaid with different temperature levels. These aspects lead to high standards for the test specimens and measuring devices.

In literature different loading systems for performing simple shear measurements are listed. For example, single-lap shear specimen (also called: single-lap joint geometry, cf. [17,18]), cylindrical double shear testpiece (cf. [1,4]) and rail shear specimen (test fixtures with bolts, cf. [22,26]). These loading systems have different advantages and disadvantages. Especially for tests in an extended temperature range a loading system without bonding/vulcanization is required. In [10] as well as [11] a new shear device with pins is presented, which enables precise shear measurements using a form fit connection.

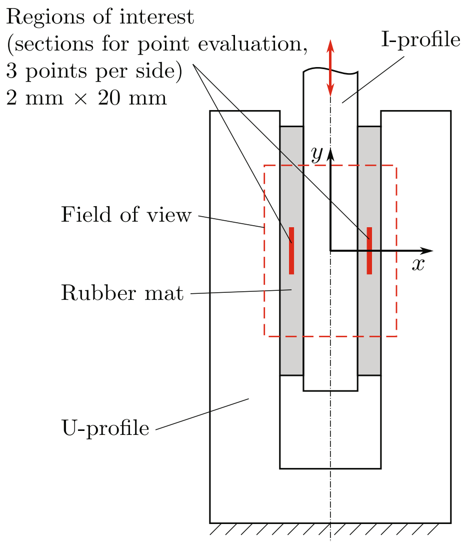

FIGURE 1 Load case and regions of interest of the characterization test, basic principle

For precise shear deformation analyses, a high-performance non-contact method is necessary. The 3D digital image correlation (DIC) fulfills the high requirements given by the characterization tests and is basically applicable at elevated temperatures (here related to rubber materials $T \in [20,100]$°C, cf. [22]). DIC basics and examples of applications are given in [6,24]. Using the primary DIC results, strain determinations can be performed by various methods, like, for example, local calculations on the basis of measured coordinates of adjacent points (cf. [7]), which can be mentioned as a standard procedure. For the purpose of smoothing, thereby the results can be processed by means of filters, like, for example, spatial mean value or median filters in the adjacent region of a point (cf. [23]). Another approach is a global method of strain calculation based on smoothing DIC displacement data using the finite element method (FEM), described in [3,24]. Furthermore, strain determination can be carried out based on global functions, for example, polynomials (cf. [12,19]) or B-splines (cf. [8] regarding approximation, [14,15,25] regarding approximation and strain evaluation), which are obtained by approximation respectively. Besides providing functions for the strain calculation, smoothing of noisy raw data is achieved using these approximation methods.

As a consequence, the new shear device with rubber mat is applied and evaluated for an extended temperature range. In the presented investigations the standard evaluation method (implemented in ARAMIS) and, for comparison as well as noise evaluation, a special approximation based method considering characteristic noise properties of the data are performed for shear strain determination.

2 | LOAD CASE AND EXPERIMENTAL SETUP

In the following, the load case, the shear device with pins, the rubber mat, the measuring device as well as the regions of interest are introduced. The shear loading of the rubber mat is based on the relative displacement of an I-profile to a fixed U-profile, where the specimen is fixed by pins within the device. The principle of the load case and the loading device is given in Figure 1. Furthermore, the regions of interest for point and surface evaluations are illustrated. Note that adequate evaluation is not limited to the center of the specimen height, which is caused by the good uniformity of load transfer by the pins. Hence, the determination of deformation values could be performed in sections closer to the upper and lower edge as well.

In Figure 2, the assembly of the shear device consisting of the U-profile, the I-profile, adapters, and pins are depicted. A detailed description of the device and the rubber mat preparation (punching process and DIC coating) can be found in the work of [11]. The shear device itself is installed in a testing machine from ZWICK/ROELL, see Figure 2. The force is measured with a load cell (nominal load 20 kN) and the deformation analysis is performed with a 3D DIC measuring system (GOM ARAMIS 4M, hardware parameters cf. Table A1 in the Appendix). Moreover, an aiming kit is used to minimize offsets and angular misalignments. In Figure 2, also the temperature chamber for measurements in an extended

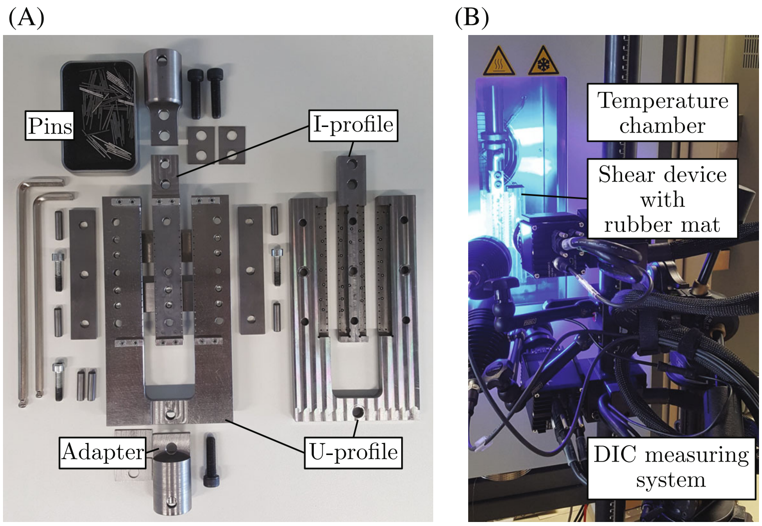

FIGURE 2 (A) Assembly of the shear device, (B) experimental setup with measuring device

temperature range is depicted. Note that the temperature chamber has an inspection glass with double glazing. For the later DIC measurements, tests without and with double glazing are performed as reference tests (see Section 4). As a result, the measurement error resulting from double glazing as well as the evaluation method can be validated. For this, the standard evaluation method from GOM ARAMIS will be compared with a special approximation based evaluation and smoothing method. DIC analysis parameters including, for example, subset size (facet size) and step size (distance between the DIC identification points) are given in Table A2 in the Appendix. Typically, larger subset sizes result in lower displacement noise (but with spatial smoothing effect), while for lower facet sizes the noise increases (cf. [9]). The used subset size of 19 pixels is the standard value in the software ARAMIS, which leads to adequate properties regarding noise and consideration of spatial inhomogeneities. Regarding the fineness of the speckle pattern, the measuring volume used (respectively the field of view) must be taken into account. The noise-floor (cf. also Table A2) is computed exemplarily for room temperature conditions via the standard deviation of the displacement/strain field for 20 different images in the reference state (without force and displacement). For the two-sided stepped shear tests a traverse speed of $v = 80$ mm/ min and a maximum shear value of approximately $s = 1.5$ was applied. For each load level three cycles were driven and 4 Hz was selected as measuring frequency.

3 | EVALUATION METHOD

3.1 | Definition of measures, standard ARAMIS method

In this section, the basics of stress and shear value calculation according to the load case given in Figure 1 are described. The stress T_{xy} (1. Piola Kirchhoff stress) is defined as the shear force F_y divided by the initial area A_0 (determined by the specimen dimensions):

$$T_{xy} = F_y / A_0 \ . \tag{1}$$

The shear value s is specified as the tangent function of the shear angle θ_{xy} (cf. [18]):

$$s = \tan\left(\theta_{xy}\right) \ . \tag{2}$$

In Figure 3, a uniform cube in the reference configuration $\tilde{K}$ and the deformed configuration K is illustrated for a shear value of $s > 1$. For the later experimental studies (see Sections 4 and 5) evaluated with the standard ARAMIS method, the shear angle is measured optically regarding local tangential directions (implemented in DIC software ARAMIS Professional 2020, cf. [7]) and calculated to a shear value. The DIC facet matching algorithm was carried out step by step.

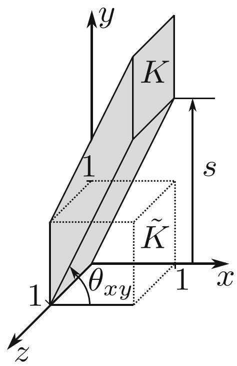

FIGURE 3 Simple shear deformation of a uniform cube with a shear value of $s > 1$

For the determination of the characterization results six evaluation points (three left and three right) are used. The sign of the shear value corresponds to that of the left region of interest. Thus, the shear value of the right section generally is multiplied with −1. Finally, the results of the left and the right points are averaged to a global shear value.

3.2 | Approximation based method

Next, the deformation evaluation based on spline approximation is described. This procedure mainly addresses strain field evaluations. By using defined points of the obtained field information, a comparison with the described standard ARAMIS method can be performed. The new approximation based method enables subtle variation and adjustment of raw data smoothing as well as an evaluation of noise error influences.

The 3D coordinates as primary DIC results are further processed and used for determining the shear strain measures. Due to the noise of the raw DIC data, the 3D coordinates are approximated by B-spline surfaces (using the MATLAB curve fitting toolbox), which is basically described, for example, in [8]. Furthermore, in this way, functions for the subsequent evaluation are provided in order to calculate various tensorial strain measures. In the analyses of this contribution cubic B-splines were used. The spline approximation is mainly controlled by the smoothing parameter—tolerance (Tol_a, with the 3D coordinates a). In addition, local differences in the smoothing properties can be taken into account by defined weighting parameters. Examples for application of an approximation based method with manually adapted tolerance parameters considering smoothing of noise are given, for example, in [14,15,25], including strain calculation.

As a further development of such approximation procedures, the tolerances are now to be optimized. In Figure 4 the basic principle of the new method is shown, which is described in the following. Fundamentals of the method are also presented in Lehmann and Ihlemann [16]. The basic idea is to consider characteristic noise properties of the measuring results during the approximation and smoothing procedure. This is intended to improve smoothing of noise and the preservation of true gradients in the distributions. For this purpose, the reference state is analyzed by the calculation of the noise-induced shift (a kind of apparent displacement) of the 3D coordinates (DIC data) using an acquired image sequence (minimum of 2 images). This results in the matrix $[\Delta\tilde{a}_{ij}]$. For comparison, the resulting noise in the deformed state is determined by the difference of the DIC and the smoothing spline coordinate values in the form of matrix $[\Delta a_{ij}]$. On this basis, the similarity of characteristic noise properties in the reference and deformed state is evaluated by means of 2D fast Fourier transforms (FFT) (cf. [5]) in MATLAB. Thus, the quantification and the assessment of the "waviness" of the 3D coordinate fields is enabled. For suitable similarity results the 2D FFT of the specific matrices is performed twice.

First, a 2D FFT of the obtained difference matrices (noise $[\Delta\tilde{a}_{ij}]$ and $[\Delta a_{ij}]$) is carried out, followed by the determination of the magnitudes (matrices $[\bar{A}^a_{ij}]$ – reference state, $[A^a_{ij}]$ – deformed state). Subsequently, a second 2D FFT of these calculated matrices is performed, likewise with calculation of the magnitudes (matrices $[\bar{B}^a_{ij}]$ – reference state, $[B^a_{ij}]$ – deformed state). For the smoothing process a good correlation of the noise characteristic in the reference and the deformed state is aspired with adaption of the tolerance Tol_a. Hence, the Pearson correlation coefficient C (cf. [20]) is used as a similarity measure, where the theoretical maximum value of C equals to 1 (which would mean a complete correlation). On this basis, an optimization by means of the Nelder-Mead-Simplex algorithm (using MATLAB) is performed minimizing the following object function to obtain suitable tolerances (adapted during optimization) for the

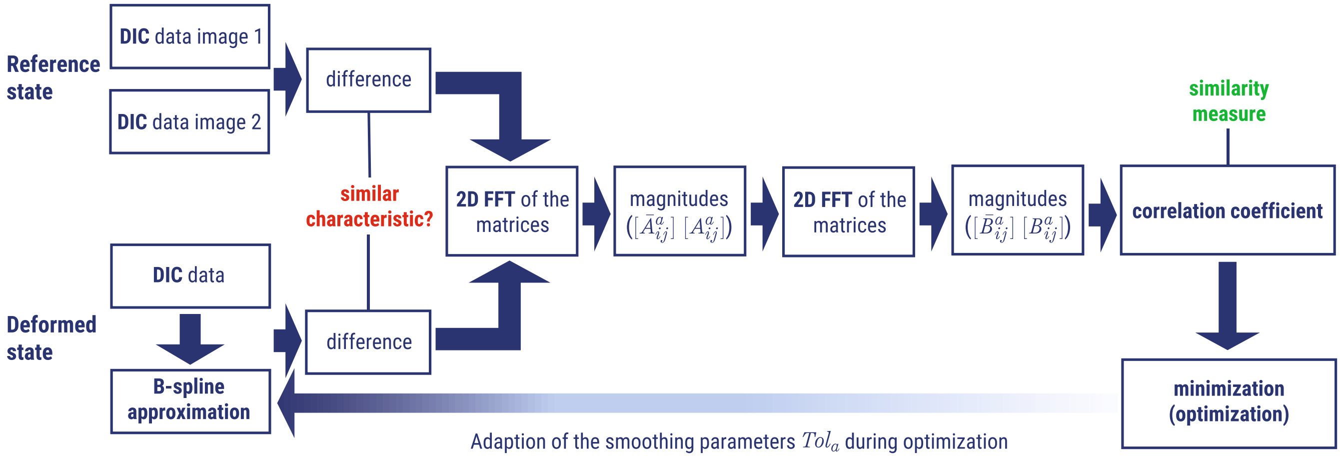

FIGURE 4 Principle of the new evaluation method using B-spline approximation of the 3D coordinates considering characteristic noise properties

approximation procedure:

$$\min_{Tol_a}\left(1 - C([\bar{B}^a_{ij}], [B^a_{ij}(Tol_a)])\right). \tag{3}$$

Based on the obtained approximated 3D coordinates (spline surfaces), the deformation gradient $\underline{\underline{F}}$ regarding local tangential directions $\tilde{x}, \tilde{y}$ (reference state) and x, y (deformed state) on the specimen surface is calculated. The Green-Lagrange strain tensor than is determined by:

$$\underline{\underline{\gamma}} = \frac{1}{2}\left(\underline{\underline{F}}^{\mathrm{T}} \cdot \underline{\underline{F}} - \underline{\underline{I}}\right) \tag{4}$$

with $\underline{\underline{I}}$ as the identity tensor.

By using the coefficients of the Green-Lagrange strain tensor the shear angle θ_{xy} is defined by:

$$\theta_{xy} = \arcsin\left(\frac{2\gamma_{xy}}{\sqrt{1+2\gamma_{xx}}\,\sqrt{1+2\gamma_{yy}}}\right). \tag{5}$$

Finally, for the evaluations presented in this article the shear value s is calculated using Equation (2).

4 | REFERENCE MEASUREMENTS AT ROOM TEMPERATURE

The shear tests presented hereafter were performed with filled ethylene-propylene-diene monomer (filled EPDM) rubber. First of all, reference measurements without and with temperature chamber were carried out to check the influence of double glazing (cf. Figure 2) on the DIC measurement. Note that double glazing can lead to undesirable effects such as reflection and/or measurement blur. In Figure 5 the measurements without and with double glazing are depicted. For better comparability, the rubber mat was softened in a first test run. As can be seen from the diagram, the tests without and with double glazing lead to almost the same stress–shear value curves. This result is insofar very important, because the later temperature tests can only be carried out with the help of the temperature chamber.

Next, the out-of-plane component to the rubber mat will be analyzed in more detail. For this purpose, a non-dimensional measure κ is introduced, which is based on the maximum and the minimum out-of-plane components in the measuring zone as well as the specimen thickness t_0:

$$\kappa := \left|u_{z,\max} - u_{z,\min}\right| / t_0 \,. \tag{6}$$

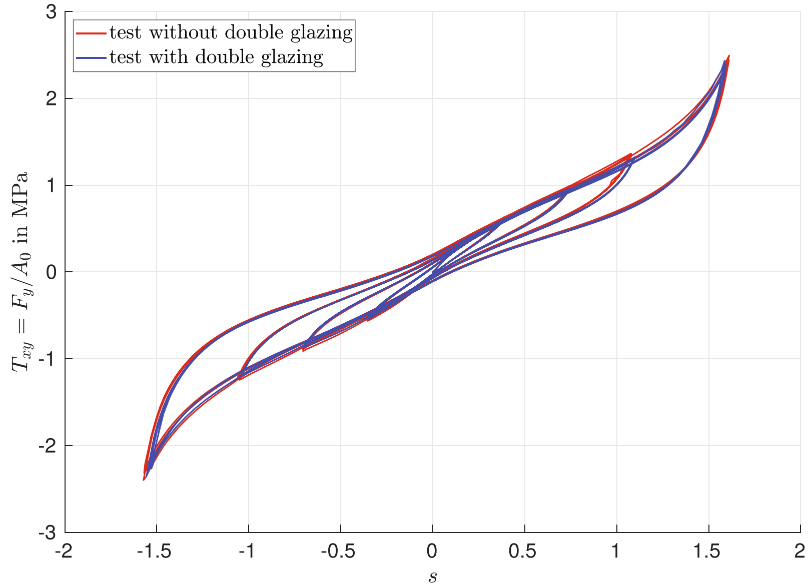

FIGURE 5 Stress–shear value diagram for tests without and with double-glazing (measurements for filled EPDM rubber)

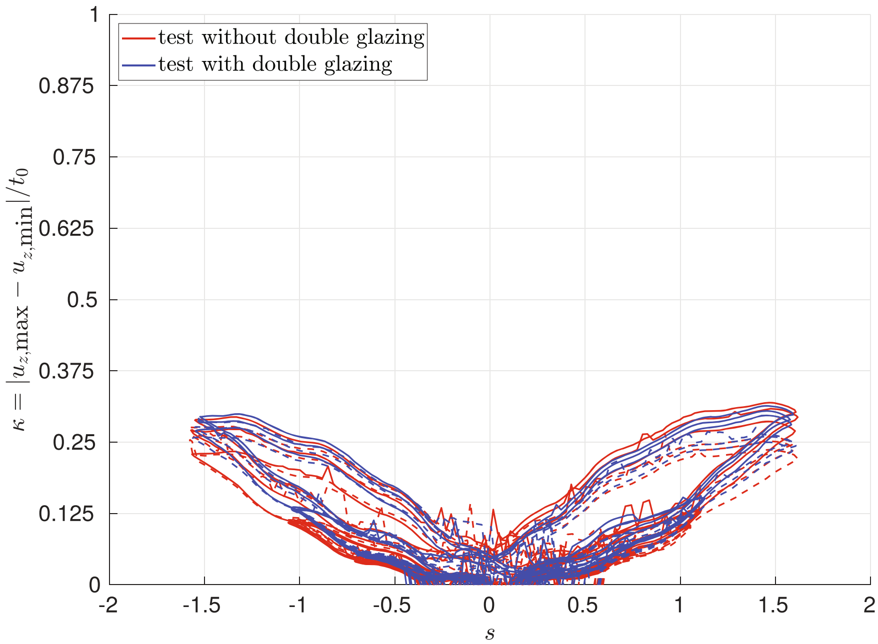

FIGURE 6 Kappa–shear value diagram for tests without and with double-glazing (measurements for filled EPDM rubber)

In Figure 6, the instability measure κ versus the shear value s is depicted for the two shear tests. It should be pointed out that the instability measure κ is very small for all shear tests. As a consequence, there is no risk of buckling. For more information about numerical studies of buckling and experimental validation of buckling phenomena see [11].

In the following, the specimen variance will be investigated in more detail. For this reason, rubber mats with different thicknesses as well as rubber mats with the same thickness are compared with each other. Note that the shear device with pins can be applied modularly for various material thicknesses. In Figure 7, the stress–shear value curves are depicted for different specimens. As a result, a very good correlation of the stress–shear value curves can be observed. All specimens lead to almost identical material behavior. As a consequence, the influence of specimen variance can be neglected. Furthermore, typical rubber properties like hysteresis, softening and Mullins effect can be identified.

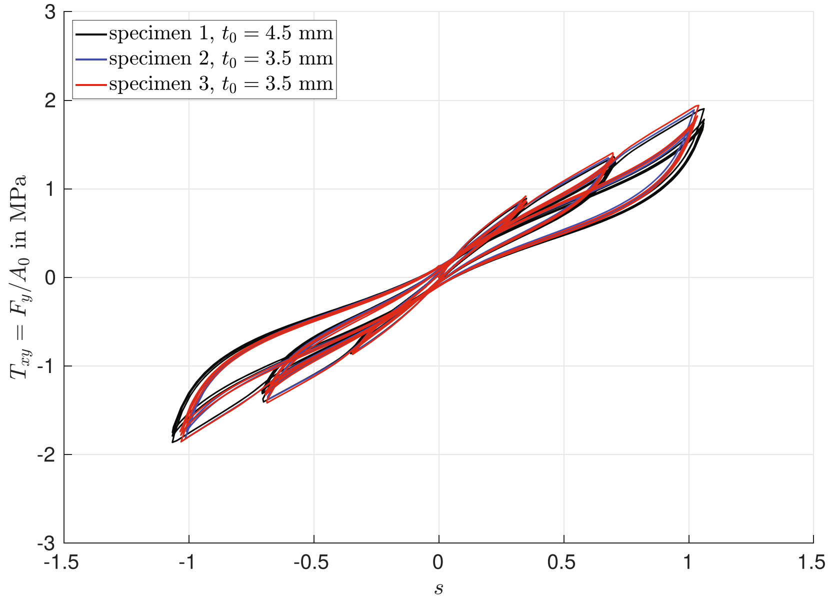

FIGURE 7 Stress–shear value diagram for different rubber mats (measurements for filled EPDM rubber at room temperature)

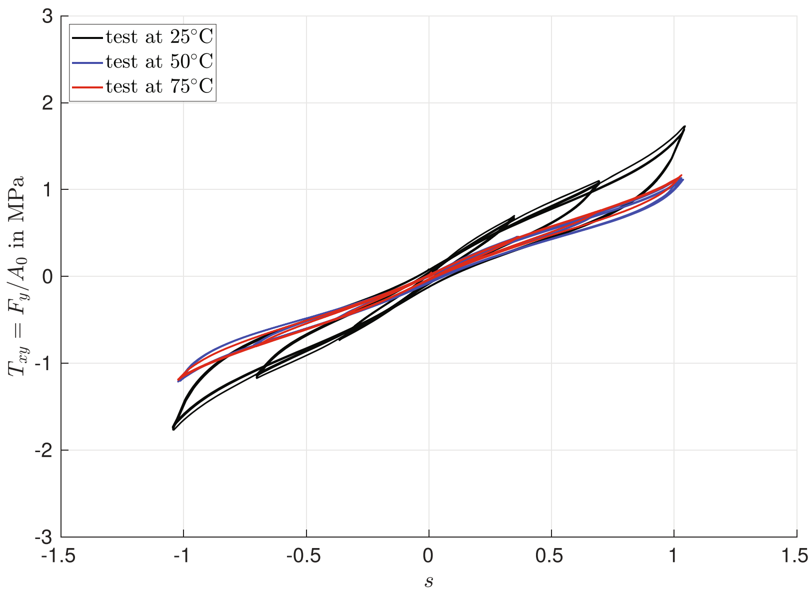

FIGURE 8 Stress–shear value diagram for various temperature levels (measurements for filled EPDM rubber)

5 | SHEAR MEASUREMENTS IN AN EXTENDED TEMPERATURE RANGE

After these basic studies, the influence of temperature to the material behavior of filled EPDM rubber will be investigated. For this purpose, two-sided stepped shear tests were conducted for room temperature (25°C), for 50°C and 75°C. In Figure 8, the corresponding stress–shear value curves are illustrated. In principle, it can be confirmed that the new shear device with pins enables shear measurements at higher temperatures. Note that after disassembly, no damage or crack initiation of the rubber mat was detectable especially for the punched holes. Based on the measurement data, a strong temperature dependency of filled EPDM rubber can be obtained between room temperature and 50°C. As a consequence, the hysteresis area is decreasing significantly with increasing temperature. For a temperature of 75°C the hysteresis area decreases only slightly. It is remarkable that the DIC measurement can be applied for such high shear and temperature levels (tests with double glazing).

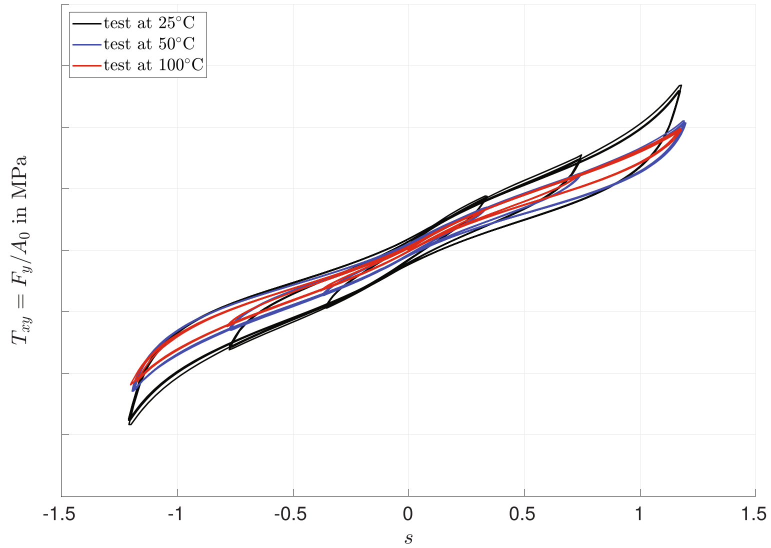

FIGURE 9 Stress–shear value diagram for various temperature levels (measurements for a secret rubber compound)

In addition, the characterization method was used for another rubber compound [1], which is designed for higher application temperatures. In Figure 9, the stress–shear value curves are depicted. Here, the decrease of the hysteresis area is not as strong as in the case of the filled EPDM rubber. Also for an application temperature of 100°C viscoelastic properties can be detected.

6 | RESULTS OF THE APPROXIMATION BASED EVALUATION METHOD

In this section, selected states of a subset of tests are evaluated using the special approximation based evaluation method described in Section 3. The aim is the characterization of noise properties and their influence as well as the comparison with the standard ARAMIS method used for the stress–shear value diagrams of Sections 4 and 5.

Applying the procedure with optimization of the respective tolerance parameters Tol_a, the results show good correlation of the noise characteristic in the reference and in the deformed state. Regarding the entire analyses, in a few cases it was necessary to use the tolerance parameter of a selected suitable local minimum of the object function calculated during the optimization process to obtain a plausible smoothing result. In Figure 10 an example of coordinate noise distributions (the differences $\Delta\tilde{x}$ (reference state) and Δx (deformed state)) for tests at room temperature in the left region of interest is illustrated. The deformed states are depicted for the first cycles (load level 1) of the tests with shear value $s \approx -0.35$. Both cases—without and with double glazing—show similar noise characteristic and range, which underlines the conclusion, that the double glazing does not influence the measurements significantly.

Furthermore, result examples of evaluated distributions of the shear value s at room temperature in the region of interest are demonstrated in Figure 11. Therein, the tests without and with double glazing are compared appropriate to Figure 10. For the test with double glazing, the results of three different load levels are shown. According the method, these plots show already smoothed distributions. Generally it can be concluded that inhomogeneities occur (increasing with rising load), but overall with a small amount. These weak inhomogeneities are caused by the pins of the device and by possible superimposed accumulated errors at the higher load levels due to the DIC algorithm (step by step analysis), which are not eliminated by the presented smoothing procedure (considering the noise characteristic in the reference state). The comparison of the results without and with double glazing show good correlation and confirm the conclusions of the noise considerations given above.

For a detailed analysis and comparison of the evaluation methods a calculation of shear values is performed along a path in y-direction close to the point evaluation positions, which is depicted in Figure 12. The comparison is carried out for the special approximation based method with optimization according to Figure 4, without optimization (that

[1] For reasons of confidentiality, no components can be specified.

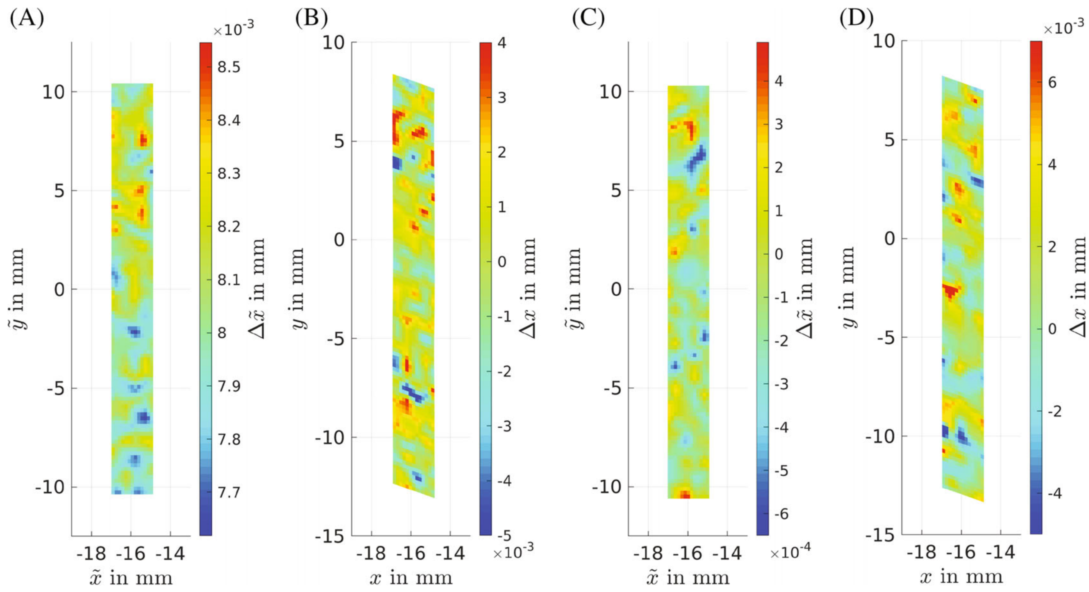

FIGURE 10 Noise distributions (of the coordinate x), tests at room temperature, data according to the tests shown in Figure 5, left region of interest: (A) without double glazing—reference state, (B) without double glazing—deformed state (load level 1), (C) with double glazing—reference state, (D) with double glazing—deformed state (load level 1)

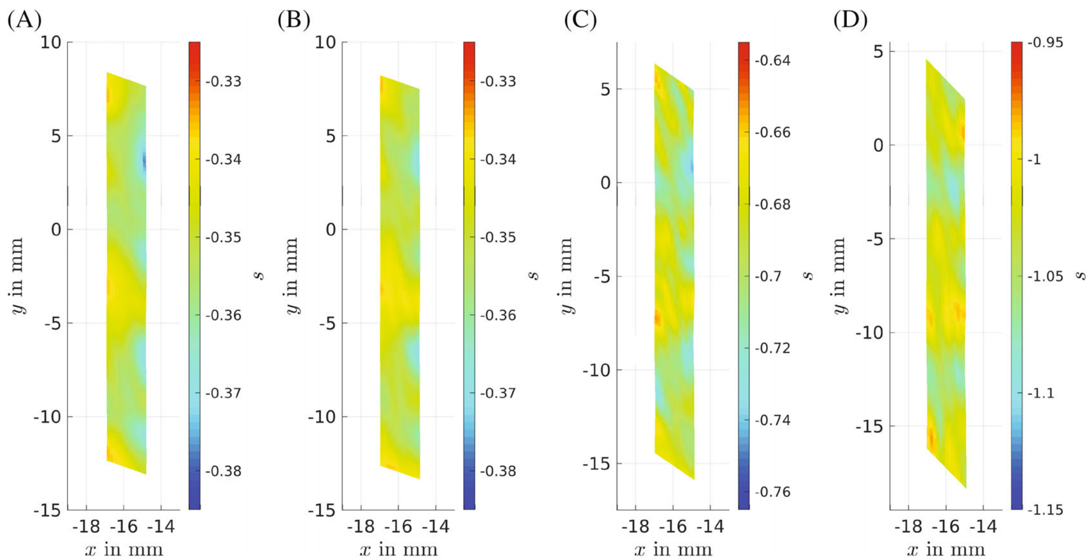

FIGURE 11 Distributions of the shear value s, tests at room temperature, data according to the tests shown in Figure 5, left region of interest: (A) without double glazing—load level 1, (B) with double glazing—load level 1, (C) with double glazing—load level 2, (D) with double glazing—load level 3

means the tolerances Tol_a are manually set to a very small value) and the standard ARAMIS method. Too small tolerance values lead to nearly unsmoothed curves of the shear values showing a range along the path of $\Delta s \approx 0.06$. Hence, these tolerance values should not be used for the characterization evaluations. By optimizing the tolerances considering the noise properties, a more plausible and well smoothed course of the shear value curve is obtained (range along the path of $\Delta s \approx 0.02$). The standard ARAMIS method shows more oscillations than the smoothed curve but provides suitable values for the evaluation within the framework of characterization (range along the path of $\Delta s \approx 0.03$). A certain smoothing effect is already achieved by the set step size (cf. Table A2 in the Appendix) of the ARAMIS settings.

Moreover, in order to assess the precision and plausibility of the characterization test results, shear values of the six used evaluation points are determined using the presented approximation based method and exemplarily compared with

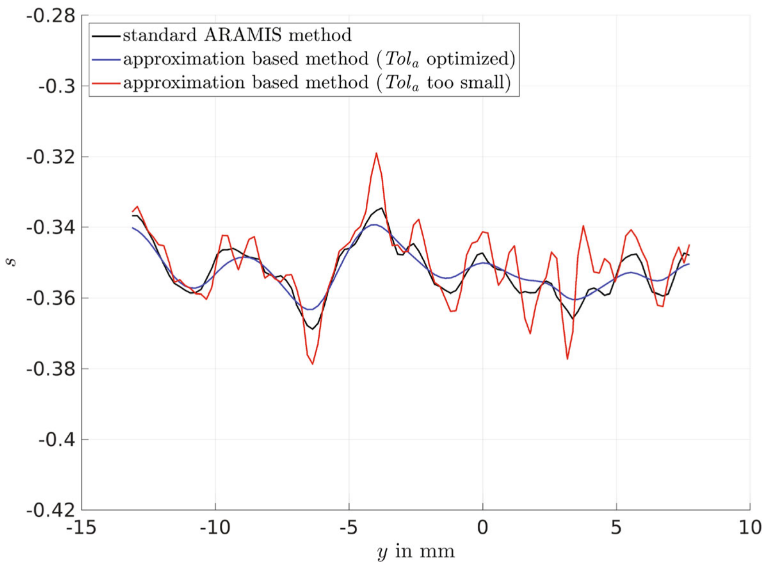

FIGURE 12 Shear values s along a path close to the point evaluation positions, test with double glazing at room temperature, data according to the tests shown in Figure 5, load level 1, left region of interest, comparison of different evaluation methods

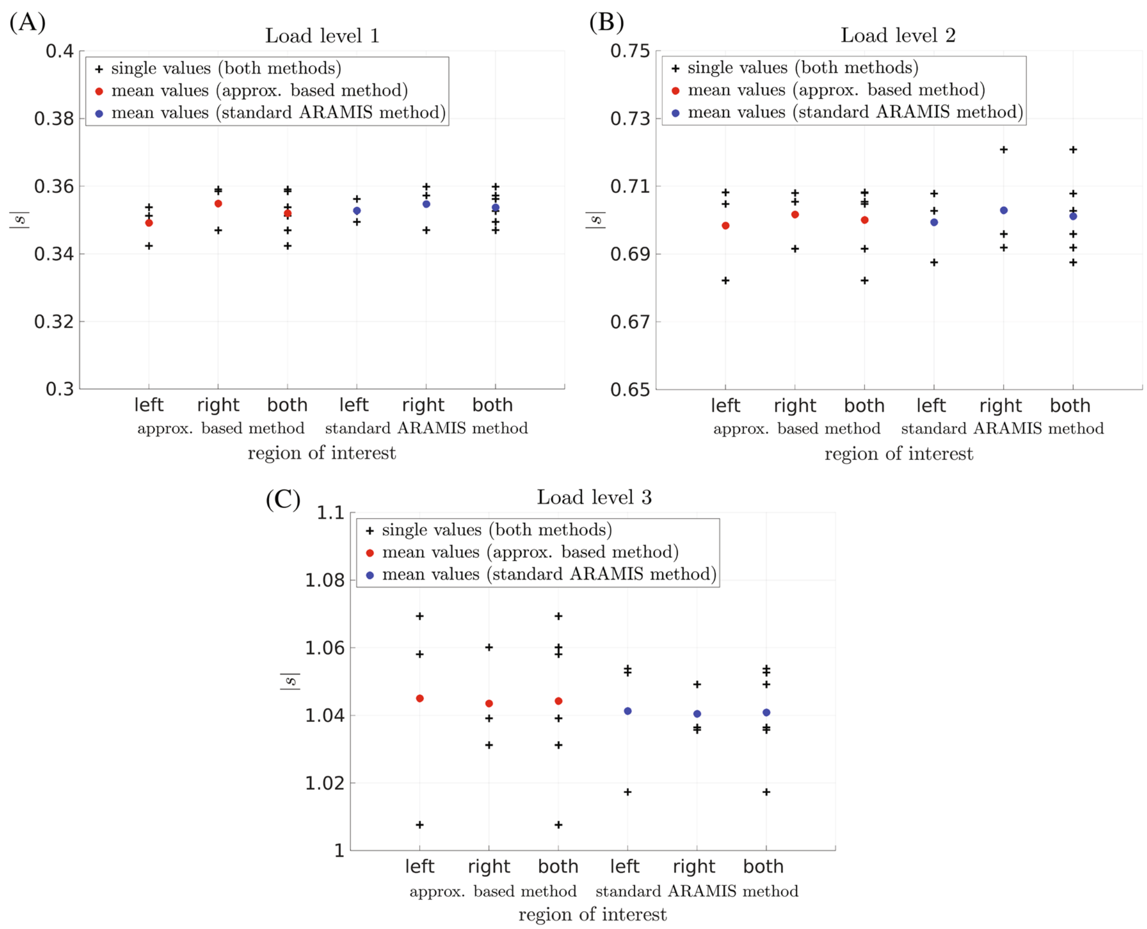

FIGURE 13 Absolute value of shear value $|s|$—Point evaluation of the regions of interest, comparison of the approximation based method with standard ARAMIS method, test with double glazing at room temperature, data according to the tests shown in Figure 5: (A) load level 1, (B) load level 2, (C) load level 3

results of the standard ARAMIS method. In Figure 13, the point results in form of single and mean values in the left and the right region of interest are given for a test at room temperature with double glazing according to the data shown in Figure 5. Additionally, the averaged values for both sides are depicted. It is demonstrated that deviations between the left and the right region of interest are generally in a small level. Furthermore, deviations of the single values from mean value increase with rising deformation, which corresponds to the results of the shear value distributions given in Figure 11. Moreover, the results obtained by the special approximation based method and the standard ARAMIS method show good correlation, especially for the load level 1. This leads to the conclusion, that the standard ARAMIS method works well for the given purpose of characterization due to averaging the six point values.

Additional results of point evaluation using the approximation based method are presented in the Appendix B. In Figure B1A the shear value results of a test without double glazing at room temperature and in Figure B1B the results of a test with double glazing at 75°C are depicted for the load level 1. The comparison of both test results shows that the deviation between the left and the right side and differences between the single and mean values are very similar. Both examples show a slightly higher level of deviations than the test with double glazing at room temperature given in Figure 13.

7 | CONCLUSIONS

In this contribution a new shear specimen (consisting of shear device with rubber mat and pins) was applied for measurements in an extended temperature range. First of all, it was checked how large the deviations are for measurements without and with double glazing. Figure 5 shows that no significant deviations occur. Furthermore, it was verified how large the scatter is for different rubber specimens. As a result, a very good correlation of the stress–shear value curves can be observed (cf. Figure 7). Next, shear tests were carried out for different temperature levels. Thereby, it can be confirmed that with the loading system of a form fit connection such measurements can be conducted. In comparison to this, problems occur by bonding or vulcanizing. Figures 8 and 9 shows the stress–shear value curves for various temperature levels. As a result, it was found, that the stresses as well as the hysteresis width decrease with increasing temperature. These shear measurements are well suited for parameter fitting of constitutive rubber models.

Furthermore, a new approximation based evaluation method for deformation analyses was performed with consideration of characteristic noise properties. This provides functions for further processing (determination) of shear strain values and smoothing of the raw DIC data. The approximation of the 3D coordinates was carried out using B-spline surfaces. Based on the initial noise characteristic (noise in the reference state—with no applied force and displacement) the B-spline approximation was performed with optimized controlling (tolerance) parameters for smoothing. This provided an assessment of noise influences on the measurement and a comparison with the standard ARAMIS method for strain value calculation. The analyses show no significant influence of the double glazing on noise characteristic and range (cf. Figure 10). Generally a low inhomogeneity level of the shear value occurs in the region of interest for the evaluation. Thus, errors due to inhomogeneities are reduced to a suitable low level for the demonstrated characterization investigations. At larger deformations inhomogeneities of the shear value distribution on the specimen surface increase due to the influence of the shear device pins and accumulated errors (cf. Figure 11). Furthermore, it can be concluded that the standard ARAMIS method with the use of 6 evaluation points is suitable for the shear value determination, which was proved by the approximation based method. In comparison, the standard ARAMIS method shows more oscillations of the shear value distributions, but in a range that has no significant influence on the characterization results (cf. Figure 12). Moreover, the deviations between the single and mean values of the point evaluation (basis for the shear value used for the characterization results) are low and approximately in the same range for both evaluation methods with an increasing amount by rising load level (cf. Figures 13 and B1).

As a consequence, the new shear device with pins as well as the evaluation methods can be applied for high-precision shear measurements especially at higher temperatures. In addition, also other rubber-like materials (e.g., thermoplastic elastomers) can be tested and characterized with the described procedure.

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APPENDIX A. TABLES

The tables satisfy the reporting requirements of the *Good Practices Guide for Digital Image Correlation* (cf. [9]).

TABLE A1 DIC hardware parameters

Camera	TELEDYNE DALSA (ARAMIS 4M system)
Image resolution	2358 × 1728 pixels^2
Lens	Titanar 50 mm
Aperture	f/11
Field-of-view	80 × 60 mm^2
Image scale	29.5 pixels/mm
Stereo-angle	25 degrees
Stand-off distance	395 mm
Image acquisition rate	4 Hz
Patterning technique	white spray paint
Pattern feature size (approximate)	4 pixel/0.14 mm

TABLE A2 DIC analysis parameters (strain data refers to standard ARAMIS evaluation method)

DIC software	ARAMIS Professional 2020
Image filtering	No filtering
Subset size	19 pixels/0.64 mm
Step size	16 pixels/0.54 mm
Subset shape function	Pseudo-affine
Matching criterion	Sum of square differences (SSD)
Interpolant	Bilinear
Strain window	7 data points
Virtual strain gauge size	83 pixels/2.8 mm
Strain formulation	Shear angle
Post-filtering of strains	No filtering
Displacement noise-floor	0.01 pixels/0.35μm (in-plane); 0.04 pixels/1.5μm (out-of-plane)
Strain noise-floor (ε_{xy})	0.0002

APPENDIX B. ADDITIONAL FIGURES

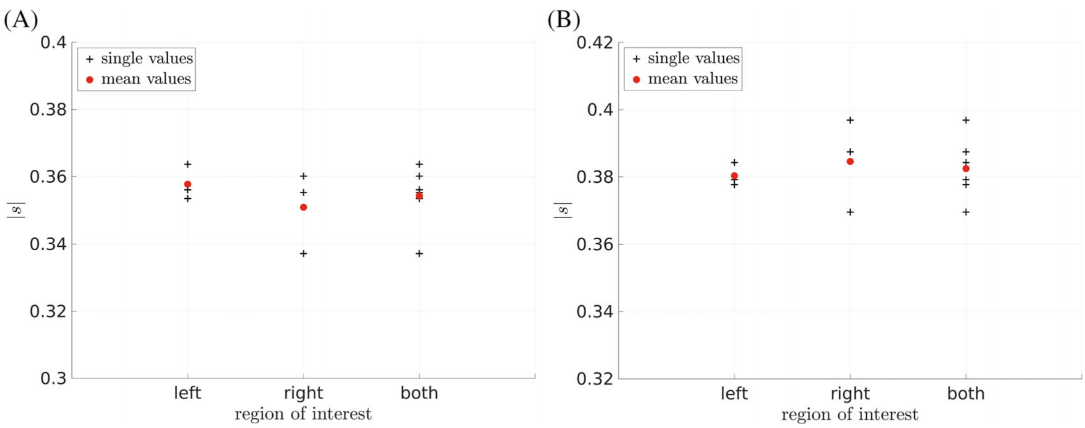

FIGURE B1 Absolute value of shear value $|s|$—Point evaluation of the regions of interest, approximation based method, load level 1: (A) test without double glazing at room temperature—data according to the tests shown in Figure 5, (B) test with double glazing at 75°C